Pranav Nagarajan

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Education

California Institute of Technology — Pasadena, CA
Ph.D. in Astrophysics — *Expected 2027*
M.S. in Astrophysics — *2024*
Advisor: Kareem El-Badry — GPA: 4.2/4.0

University of California, Berkeley — Berkeley, CA
B.A. in Astrophysics, Physics, and Data Science — *2022*
Highest Distinction in General Scholarship — GPA: 4.0/4.0

Research Experience

Graduate Student Researcher — California Institute of Technology
Advisor: Prof. Kareem El-Badry — *2022-Present*

- Discovering and Characterizing Stellar-Mass Black Holes and Neutron Stars in Galactic Binaries

Undergraduate Student Researcher — University of California, Berkeley
Advisor: Prof. Daniel Weisz — *2020-2022*

- Mapping the Local Group with RR Lyrae and Bayesian Hierarchical Modeling

BSRC Summer Internship — Berkeley SETI Research Center
Mentor: Dr. Vishal Gajjar — *Summer 2021*

- Breakthrough Listen Periodic Spectral Signals Search

Undergraduate Research Apprenticeship Program — Space Sciences Laboratory
Mentor: Dr. Andreas Zoglauer — *2019-2021*

- Using 3D Convolutional Neural Networks and Graph Neural Networks for Compton Track Identification in Gamma-Ray Space Telescopes

Technical Skills

- Coding: Python (pandas, Matplotlib, NumPy, SciPy, scikit-learn, Keras, TensorFlow), SQL (ADQL), Jupyter Notebook, Git, LaTeX
- Statistical Modeling: Bayesian Inference, Markov Chain Monte Carlo, Hypothesis Testing, Time Series Analysis, Machine Learning

Teaching Experience

Teaching Assistant — California Institute of Technology
High-Energy Astrophysics, Galaxies and Cosmology — *2024*

Undergraduate Student Instructor — University of California, Berkeley
Principles and Techniques of Data Science — *2020*

Honors & Awards

Neugebauer Scholar 2025
NSF Graduate Research Fellowship Program: Honorable Mention 2024
URAP Summer Award 2020
National Merit Scholarship 2018

Awarded Telescope Time

Palomar Hale 200-inch . . . . 10 nights (PI) + 3 nights (Co-I)
Keck 10m . . . . 19 nights (Co-I)
FAST 500m . . . . 3.0 hours (PI)

Observing Experience

Keck I Telescope, High Resolution Echelle Spectrometer – 4 nights 2025–2026
Palomar Hale Telescope, Next Generation Palomar Spectrograph – 4 nights 2025–2026
Keck II Telescope, Near-Infrared Camera 2 – 4.5 nights 2024–2026
Keck II Telescope, Near-Infrared Echellete Spectrometer – 3 nights 2024
Very Large Telescope, Focal Reducer and Low Dispersion Spectrograph 2 – 1 night 2024
Magellan I Telescope, Inamori Magellan Areal Camera and Spectrograph – 1 night 2024
Palomar Hale Telescope, Wide-Field Infrared Camera – 2 nights 2024
Keck I Telescope, Low Resolution Imaging Spectrometer – 1 night 2024
Palomar Hale Telescope, Caltech High-speed Multi-color Camera – 1 night 2023
Palomar Hale Telescope, Double Spectrograph – 3.5 nights 2023
Keck II Telescope, Echellete Spectrograph and Imager – 1 night 2022

Journal Referee

MNRAS, ApJ, AJ, OJAp, ApJL, New Astron., Sci. Bull. . . . . 10 papers total, 2023 – Present

Selected Talks

"Black Holes from *Gaia*: Observational Tests of Formation Models,"
UCSD STRAND Talk, San Diego, CA . . . . May 2026

"The Puzzle of PSR J1928+1815: An Eclipsing MSP with a Mysterious Companion,"
CIERA-Northwestern, Evanston, IL . . . . May 2026

"A Spectroscopic Search for Dormant Black Holes in Low-Metallicity Binaries,"
Harvard CfA Institute for Theory and Computation, Cambridge, MA . . . . Mar. 2026

"Observational Tests of Formation Models for Gaia BHs,"
MIT Kavli Institute, Cambridge, MA . . . . Mar. 2026
"A Spectroscopic Search for Dormant Black Holes in Low-Metallicity Binaries,"
The Lifecycle of Stellar Black Holes conference, Santa Barbara, CA . . . . Nov. 2025
"Observational Tests of Formation Models for Black Holes in Binaries,"
Kavli Institute for Theoretical Physics, Santa Barbara, CA . . . . Nov. 2025
"Searching for Tertiaries to Black Hole and Neutron Star Binaries with NIRC2-LGS,"
Keck Science Meeting 2025, Los Angeles, CA . . . . Sep. 2025
"Realistic predictions for Gaia BH discoveries,"
Binary Stars in the Space Era conference, Keele, United Kingdom . . . . Jul. 2025
"The SyXB IGR J16194-2810: A Window on the Future Evolution of Gaia NSs,"
Symbiotic stars conference, Prague, Czech Republic . . . . Jun. 2024
"The SyXB IGR J16194-2810: A Window on the Future Evolution of Gaia NSs,"
Observatories of the Carnegie Institution for Science, Pasadena, CA . . . . May 2024
"Spectroscopic follow-up of BH and NS candidates in ellipsoidal variables from *Gaia* DR3,"
Zwicky Transient Facility Team Meeting, Pasadena, CA . . . . Oct. 2023

Refereed Publications

h-index: 10 (all papers), 8 (first-author papers)

24. El-Badry, K., Latham, D. W., Rix, H.-W., Bieryla, A., Buchhave, L. A., Shahaf, S., Mazeh, T., Müller-Horn, J., **Nagarajan, P.**, Yamaguchi, N., Lam, C. Y., Rowan, D., Simon, J. D., Lu, J. R., and Isaacson, H. (2026). Spectroscopic Follow-up of Compact Object Binary Candidates from Gaia DR3: White Dwarfs, Neutron Stars, Black Holes, and the Parallax Zeropoint. Submitted to *The Open Journal of Astrophysics*, arXiv:2608.06453.

23. Müller-Horn, J., El-Badry, K., Sander, A. A. C., Rix, H.-W., Blomberg, L., Hermes, J. J., **Nagarajan, P.**, Shahaf, S., Jin, H., Rowan, D. M., Dutta, D., Fernández-Trincado, J. G., Götberg, Y., Ilyin, I., Maccarone, T., Méndez Delgado, J. E., Stringfellow, G. S., Tkachenko, A., Villaseñor, J. I., and Zari, E. (2026). A Galactic Intermediate-Mass Stripped Star with a Wolf–Rayet-like Wind. Submitted to *Astronomy & Astrophysics*, arXiv:2608.05276.

22. Schiebelbein-Zwack, A., Ye, C. S., Reina-Campos, M., Munawwarah, A., El-Badry, K., and **Nagarajan, P.** (2026). The Contribution of Disrupted Dense Star Clusters to Gaia's Compact Object Binaries. Submitted to *The Astrophysical Journal*, arXiv:2606.18419.

21. El-Badry, K., Werner, K., Shen, K. J., Strader, J., Rodriguez, A. C., Han, J. J., Chandra, V., Chomiuk, L., Vanderbosch, Z. P., Blomberg, L., Yamaguchi, N., **Nagarajan, P.**, Caiazzo, I., van Roestel, J., Glanz, H., Wong, T. L. S., Bhat, A., Hollands, M. A., and Gänsicke, B. T. (2026). A Systematic Survey for Hypervelocity Runaways from Thermonuclear Supernovae. *The Open Journal of Astrophysics, 9.*

20. van Son, L. A. C., Yamaguchi, N., **Nagarajan, P.**, Shenar, T., Sen, K., Laroche, A., Leiner, E. M., Sana, H., and Pols, O. R. (2026). Ongoing and Post-Mass-Transfer Binaries: A Living Catalog and Unified Review of Binary Mass Transfer Products. Submitted to *The Astrophysical Journal Supplement Series*, arXiv:2605.31290.

19. **Nagarajan, P.**, El-Badry, K., and Sanghi, A. (2026). Searching for the Third Wheel: High-Contrast Imaging Constraints on Tertiaries to Black Hole and Neutron Star Binaries. *Publications of the Astronomical Society of the Pacific, 138(7), 074204.*

18. **Nagarajan, P.**, El-Badry, K., Fuller, J., Guo, Y., and Tauris, T. M. (2026). Deep Adaptive Optics Imaging Rules Out a Helium Star Companion to PSR J1928+1815. *Publications of the Astronomical Society of the Pacific, 138(6), 064202.*

17. **Nagarajan, P.**, El-Badry, K., Maccarone, T. J., Iorio, G., Rastello, S., and Müller-Horn, J. (2026). BE Lyncis is not a Black Hole Binary: Lessons from Gaia and Hipparcos Astrometry. *Publications of the Astronomical Society of the Pacific, 138(4), 044201.*

16. Iorio, G., **Nagarajan, P.**, Bobrick, A., El-Badry, K., Pancino, E., Belokurov, V., Zhāng, H., D'Orazi, V., Mateu, C., Rastello, S., and Gieles, M. (2026). Hide and Seek with Gaia. Detectability of Predicted Thin-Disc Metal-Rich RR Lyrae Binaries in Gaia DR3 and DR4. Accepted to *Astronomy & Astrophysics*, arXiv:2603.20429.

15. **Nagarajan, P.**, El-Badry, K., Bobrick, A., Iorio, G., Molina, F., Vos, J., and Vučković, M. (2026). Testing models for fully and partially stripped low-mass stars with Gaia: Implications for hot subdwarfs, binary RR Lyrae, and black hole impostors. *Publications of the Astronomical Society of the Pacific, 138(4), 044205.*

14. Handley, L. B., Howard, A. W., Dai, F., Rubenzahl, R. A., Giacalone, S., Isaacson, H., Ong, J. M. J., Carmichael, T. W., Li, Y., Lubin, J., Premnath, P. H., Rogers, C. J., **Nagarajan, P.**, Gilbert, G. J., Fulton, B., Gibson, S. R., Roy, A., Edelstein, J., and Smith, C. (2026). The KPF-SLOPE Survey - Small, Compact Multi-Planet Systems Appear Spin-Orbit Aligned. Submitted to *The Astrophysical Journal*, arXiv:2603.23713.

13. **Nagarajan, P.**, El-Badry, K., Reggiani, H., Lam, C. Y., Simon, J. D., Müller-Horn, J., Seeburger, R., Rix, H.-W., Isaacson, H., Lu, J. R., Chandra, V., and Andrae, R. (2025). A Spectroscopic Search for Dormant Black Holes in Low-Metallicity Binaries. *Publications of the Astronomical Society of the Pacific, 137(9), 094202.*

12. Galiullin, I., Rodriguez, A. C., El-Badry, K., Caiazzo, I., Szkody, P., **Nagarajan, P.**, and Whitebook, S. (2025). Optical Spectroscopy of the Most Compact Accreting Binary Harboring a Magnetic White Dwarf and a Hydrogen-rich Donor. *The Astrophysical Journal Letters, 990(2), L57.*

11. **Nagarajan, P.**, El-Badry, K., Chawla, C., Di Carlo, N., Breivik, K., Rodriguez, C. L., Agrawal, P., Delfavero, V., & Chatterjee, S. (2025). Realistic Predictions for Gaia Black Hole Discoveries: Comparison of Isolated Binary and Dynamical Formation Models. *Publications of the Astronomical Society of the Pacific, 137(4), 044202.*

10. Hinkle, K., **Nagarajan, P.**,, Fekel, F. C., Mikołajewska, J., Straniero, O., & Muterspaugh, M. W. (2025). Binary Parameters for the Recurrent Nova T Coronae Borealis. *The Astrophysical Journal, 983(1), 76.*

9. **Nagarajan, P.** & El-Badry, K. (2025). Mixed Origins: Strong Natal Kicks for Some Black Holes and None for Others. *Publications of the Astronomical Society of the Pacific, 137(3), 034203.*

8. Blomberg, L., El-Badry K., Breivik, K., Caiazzo, I., **Nagarajan, P.**, Rodriguez, A., van Roestel, J., Vanderbosch, Z. P., & Yamaguchi, N. (2024). The Companion Mass Distribution of Post Common Envelope Hot Subdwarf Binaries: Evidence for Boosted and Disrupted Magnetic Braking? *Publications of the Astronomical Society of the Pacific, 136(12), 124201.*

7. **Nagarajan, P.** & El-Badry, K. (2024). Validation of Gaia DR3 orbital and acceleration solutions with hierarchical triples. *Publications of the Astronomical Society of the Pacific, 136(9), 094203.*

6. **Nagarajan, P.**, El-Badry, K., Lam, Y. C., & Reggiani, H. (2024). The Symbiotic X-ray Binary IGR J16194-2810: A Window on the Future Evolution of Wide Neutron Star Binaries From Gaia. *Publications of the Astronomical Society of the Pacific, 136(7), 074202.*

5. **Nagarajan, P.**, El-Badry, K., Triaud, A. H., Baycroft, T. A., Latham, D., Bieryla, A., Buchhave, L. A., Rix, H.-W., Quataert, E., Howard, A., Isaacson, H., & Hobson, M. J. (2024). ESPRESSO Observations of Gaia BH1: High-precision Orbital Constraints and no Evidence for an Inner Binary. *Publications of the Astronomical Society of the Pacific, 136(1), 014202.*

4. **Nagarajan, P.**, El-Badry, K., Rodriguez, A. C., van Roestel, J., & Roulston, B. (2023). Spectroscopic follow-up of black hole and neutron star candidates in ellipsoidal variables from *Gaia* DR3. *Monthly Notices of the Royal Astronomical Society, 524(3), 4367–4383.*

3. El-Badry, K., Shen, K. J., Chandra, V., Bauer, E. B., Fuller, J., Strader, J., Chomiuk, L., Naidu, R. P., Caiazzo, I., Rodriguez, A. C., **Nagarajan, P.**, Yamaguchi, N., Vanderbosch, Z. P., Roulston, B. R., Gänsicke, B., Han, J. J., Burdge, K. B., Filippenko, A. V., Brink, T. G., & Zheng, W. (2023). The fastest stars in the galaxy. *The Open Journal of Astrophysics, 6.*

2. Suresh, A., Gajjar, V., **Nagarajan, P.**, Sheikh, S. Z., Siemion, A. P., Lebofsky, M., MacMahon, D. H., Price, D. C., & Croft, S. (2023). A 4–8 GHz Galactic Center Search for Periodic Technosignatures. *The Astronomical Journal, 165(6), 255.*

1. **Nagarajan, P.**, Weisz, D., & El-Badry, K. (2022). RR Lyrae-based distances for 39 nearby dwarf galaxies calibrated to *Gaia* EDR3. *The Astrophysical Journal, 932(1), 19.*

Non-Refereed Publications

1. **Nagarajan, P.** (2026). Pegasi Ascendant: Ranking Constellation Genitives on their Aesthetic Merit. *arXiv e-prints,* arXiv:2604.00144.